



## 3D Printed Hexapod



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### Summary

Cheap 3D printed hexapod with lots of RGB, controlled via an app on your phone

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[hexapod](#) [stm32](#)

Seems like you want to build a Hexapod. Don't know why but since you're here, might as well tell you about the most important features about my design:

1. Cheap: 120 € (see Parts (toBuy) file)
2. RGB: 271 individual RGB LEDs
3. Cable management: Cables are routed inside the leg. no cable loops that can get tangled
4. ORP compatible: compatible with the Open Robotic Platform modules
5. Bluetooth control: Remote control via phone app

If this hasn't convinced you might I suggest watching my videos on the design and build.

Since you are reading this I assume you are at least considering my design. I have some instructions on [Instructables](#) if you want to take a look at that.

All the 3mf files are oriented correctly with the needed support material. My parts are design in a way that uses minimal support material. See "Parts (toPrint)" Excel sheet.

If you don't know what and where to buy take a look at the "Parts (toBuy)" Excel sheet. Those are the parts I bought and where. You can also update the prices and it will automatically calculate the final price. Everything I listed is without shipping since that varies drastically.

If I didn't buy it or the link doesn't work anymore I linked to where I would buy.

If you want the Code or remote App take a look at the [Github](#) or [Instructables](#) page. I have uploaded them here but the files on GitHub will be the most up to date.

To be clear, my design is NOT perfect. Here are all the problems I have:

1. highly integrated leg design: routing the cables inside the leg looks awesome but makes assembly more difficult. If you were to not use LEDs this problem would largely be negated.
2. Current spikes: the hexapod is able to handle some load but especially at times of a power spike the Buck converter powering the servos can be insufficient. So you might want to use two or buy a more powerful one.
3. I2C buss instability: I had problems with the I2C buss between the STM32 and servo drivers breaking down causing the Hexapod to freeze. I was able to fix that by using shielded cables, but it required some trial and error.
4. incomplete Code: Not all features have been added in code. For example no LED animations beside the startup animation.

Seeing the project work is the most rewarding thing for me. I know most people won't build something as complex as this but I still hope I was able to inspire you to build your own robotic Projects.

If you have any questions or suggestions consider writing me a comment, preferably on YouTube. That is the place I will most likely see it.

Happy Making :)

# Model files



## Full Hexapod Model

4 files



**hexapod-v8-rotated.3mf**



**hexapod-v8.3mf**



**hexapod-v8.f3z**

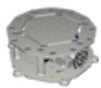


**hexapod-v8.step**

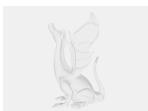


## body

13 files



**body-v35.3mf**



**body-v35.f3z**



**body-v35.step**



**pla\_body-main\_bottom.3mf**



**pla\_body-main\_bottom.stl**



**pla\_body-main\_top.3mf**



**pla\_body-main\_top.stl**



**pla\_controll-platform.3mf**



**pla\_controll-platform.stl**



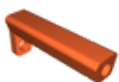
**pla\_power-pack.3mf**



**pla\_power-pack.stl**



**pla\_body-spacer.3mf**



**pla\_body-spacer.stl**

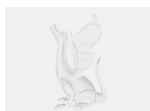


leg

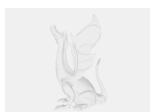
25 files



leg-v2-v20.3mf



leg-v2-v20.f3z



leg-v2-v20.step



pla\_bushing.3mf



pla\_bushing.stl



pla-coxa\_top.3mf



pla-coxa\_top.stl



pla-trochanter\_servo-mount.3mf



pla-trochanter\_servo-mount.stl



**pla-trochanter\_servo-mount\_mirrored.3mf**



**pla-trochanter\_servo-mount\_mirrored.stl**



**pla-femur\_right.3mf**



**pla-femur\_right.stl**



**pla\_femur\_left.3mf**



**pla\_femur\_left.stl**



**pla\_femur\_right\_mirrored.3mf**



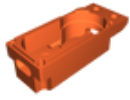
**pla\_femur\_right\_mirrored.stl**



**pla\_femur\_left\_mirrored.3mf**



**pla\_femur\_left\_mirrored.stl**



**pla-tibia\_servo-mount.3mf**



**pla-tibia\_servo-mount.stl**



**pla\_tibia\_main.3mf**



**pla\_tibia\_main.stl**



**pla\_tibia\_tip.3mf**



**pla\_tibia\_tip.stl**



**leg Test Stand**

11 files



**leg-test-stand-v4.3mf**



**leg-test-stand-v4.f3z**



**leg-test-stand-v4.step**



**test-stand\_mount.3mf**

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**test-stand\_mount.stl**

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**test-stand\_foot.3mf**

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**test-stand\_foot.stl**

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**pla-90-cuppler-40mm.3mf**

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**pla-90-cuppler-40mm.stl**

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**pla-coxa\_bottom.3mf**

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**pla-coxa\_bottom.stl**

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**leg\_set.3mf**

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**leg\_set\_mirrored.3mf**

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**pla\_power-pack.3mf**

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**pla\_body-main\_bottom.3mf**

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**pla\_body-main\_top.3mf**

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**pla\_body-lid.3mf**

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**pla\_controll-platform.3mf**

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**body-spacer\_set.3mf**

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**bushing\_set.3mf**

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## Other files

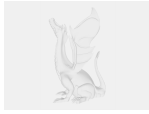


**code.zip**

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**remote.zip**



**exel.zip**

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